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Exercise:

An increase in register count, which is required to manage the segmented memory structure in SRAM, is the main source of resource consumption in the modified code. Handling separate memory regions for red, even green, odd green, even blue, and odd blue values requires the additional registers that were added. The even pixel counter, the expanded buffer registers (VGA_sram_data[5:0]), the SRAM address register, the color calculation register, and the state register, which controls the enhanced state machine, are some of the specific additions in this updated code are responsible for the increased register count. The increased register set should result in a higher overall register count in Quartus' compilation report based on the expected register usage. More address registers (such as even_pixel_counter), larger buffers (VGA_sram_data[5:0]), and control registers have been added to the state machine to support more states and enable segmented memory management.

To improve data accuracy and VGA display performance, the new SRAM memory segmentation calls for an increase in the number of registers. To facilitate efficient management of the segmented memory and satisfy the requirements of the revised architecture, the additional registers make sure that the right data is read from and written to each memory region.

Register description table:

Module (instance)	Register name	Bits	Description
Top-level experiment 1	even_pixel_counter	18	Tracks address offsets for even pixels during read and write operations, adjusting dynamically based on the segmented memory layout.
Top-level experiment 1	VGA_sram_data[5:0]	16	Buffer register - holds the Red/Green data for the even pixels before it is ready to be transferred to the VGA controller Buffers [0] and [5]: Handle red values. Buffers [1] and [4]: Store even and odd green values. Buffers [2] and [3]: Store even and odd blue values.

Top-level experiment 1	SRAM_address	18	Address register used for accessing the external memory.
Top-level experiment 1	color	3	Determines the color values for each rectangle during the initial SRAM fill operation. It ensures that the correct color values are written to the appropriate memory segments in SRAM.

Critical analysis:

When we checked the Timing Analyzer, we saw that all slack values were positive, meaning our design has no timing problems. The data arrives early enough, so there's a good safety margin. The longest delay is 12.782ns, happening from VGA_controller:VGA_unit|V_Cont[5] to SRAM_address[2]. This delay includes 7.132ns for data and -0.084ns of clock skew.